

CHEMISTRY — GRADE 10 | ACIDS AND BASES

Reading the Colours

Telling acids and bases apart with commercial indicators

Can we use colour to reliably sort the cups from the school canteen?

Seven liquids from home. Acid, base, or neutral?

- Lemon juice • Vinegar (siki) • Baking soda solution • Liquid soap solution • Tap water • Black tea (chai) • Cow's milk
- In your prediction table, write acid, base or neutral for each liquid.
- Then predict the colour each indicator will turn. Compare with your partner.

In Lesson 3, universal indicator gave us a rainbow. Will today's indicators tell the same story?

Three commercial indicators

Litmus paper

Comes in two colours, red and blue. Dip a strip and watch whether its colour flips.

Phenolphthalein

A colourless liquid indicator. Add 2 drops to the test liquid in a spotting tile.

Methyl orange

An orange liquid indicator. Add 2 drops to the test liquid in a spotting tile.

An indicator is a substance that changes colour depending on whether a solution is acidic or basic.

Before anyone opens a bottle

- Phenolphthalein is dissolved in ethanol, which burns. No open flames. Use dropper bottles only.
- Lemon juice and vinegar sting eyes. Keep hands away from your face; wash hands afterwards.
- Never taste any laboratory solution — even ones you know from home.
- Rinse used solutions down the sink with plenty of water.
- Report broken glassware. Your teacher sweeps it up, not you.

Testers test. Recorders record. Swap halfway.

- 1 Place a strip of red litmus and a strip of blue litmus on the white tile. Add 2–3 drops of the liquid to each.
- 2 Record the colour immediately and again after 30 seconds.
- 3 Put 5 drops of the liquid in a tile well. Add 2 drops of phenolphthalein. Record. Repeat with methyl orange in a new well.
- 4 Repeat for all seven liquids. Write exact colours: “turned pale pink”, not “changed”.

Be a scientist, not a poet: what exact colour do you see?

QUICK CHECK 1

Which of these is the most useful observation to record?

A “It changed.”

B “It looked interesting.”

C “Blue litmus turned bright red within 5 seconds.”

D “It was probably an acid.”

Answer key is teacher-side only, in a separate file.

What each indicator tells us

Indicator	In an ACID	In a BASE	In a NEUTRAL solution
Blue litmus	turns red	stays blue	stays blue
Red litmus	stays red	turns blue	stays red
Phenolphthalein	colourless	pink	colourless
Methyl orange	red	yellow	yellow

Write a Pattern Statement for each indicator: “An acid turns ____ from ____ to ____; a base turns it from ____ to ____.”

QUICK CHECK 2

A few drops of a liquid turn blue litmus paper red. What does this tell you?

- A** The liquid is a base
- B** The liquid is an acid
- C** The liquid is neutral
- D** The litmus paper is faulty

Answer key is teacher-side only, in a separate file.

QUICK CHECK 3

In which kind of solution does phenolphthalein turn pink?

A An acid

B A base

C A neutral solution

D Any solution containing water

Answer key is teacher-side only, in a separate file.

One indicator can't always tell you the whole story.

Methyl orange

Yellow in a base ... but also yellow in a neutral solution.
On its own, it can't tell water from soap.

Phenolphthalein

Colourless in an acid ... but also colourless in a neutral solution. On its own, it can't tell lemon juice from water.

That is why we classify using evidence from ALL the indicators together.

QUICK CHECK 4

A liquid turns methyl orange yellow. Can you be sure it is a base?

- A** Yes, yellow always means a base
- B** No, a neutral solution also turns methyl orange yellow, so test with another indicator
- C** Yes, but only if the liquid is clear
- D** No, yellow means the liquid is an acid

Answer key is teacher-side only, in a separate file.

Sorting the liquids from home

Liquid	Litmus	Phenolphthalein	Methyl orange	Class
Lemon juice	blue → red	colourless	red	Acid
Vinegar (siki)	blue → red	colourless	red	Acid
Liquid soap	red → blue	pink	yellow	Base
Baking soda solution	red → blue	faint pink	yellow	Base (weak)
Tap water	no change	colourless	yellow	Neutral
Black tea (chai)	little or no change	colourless	yellow	Weakly acidic
Cow's milk	little or no change	colourless	yellow	About neutral

Expected results. Compare with your group's data and discuss any differences — especially milk and tea.

QUICK CHECK 5

Which liquid from today would you expect to turn methyl orange red?

A Liquid soap solution

B Tap water

C Baking soda solution

D Vinegar

Answer key is teacher-side only, in a separate file.

Update your model. Add one new question.

UPDATE YOUR MODEL

Take out your model of the eight canteen cups. Label each cup ACID, BASE or NEUTRAL using today's evidence. Show that "something in the liquid" is causing the colour change. Finish: "My model still cannot explain ____."

DRIVING QUESTION BOARD

Move any earlier question that today's experiment answered into the SOLVED zone, with the evidence on the back. Add one new question, e.g. How do we measure HOW acidic something is? What do acids actually DO when they react?

From the ARES offline library



VIDEO [Video: pH Scale](#)

PhET • Partial match: introduces pH, which this unit covers in Lesson 8. Use as extension.



READING [Properties of Acids \(Flexbook\)](#)

CK-12 • Good match for properties of acids



SEARCH [Search ARES: acids, bases, indicators](#)

Opens an ARES search for related videos and notes

Links open only on the school network (ares.local). Click to open during the lesson.

5 questions asked during class today

1. Which is the most useful observation to record?
2. Blue litmus turns red. What does this tell you?
3. In which kind of solution does phenolphthalein turn pink?
4. Methyl orange turns yellow. Can you be sure it is a base?
5. Which liquid would turn methyl orange red?

Ask your teacher how to submit your answers today (for example, on a paper answer slip).